

AN IMPLICIT GRADIENT METHOD FOR CONSTRAINED BILEVEL PROBLEMS USING BARRIER APPROXIMATION

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ABSTRACT

In this work, we propose algorithms for solving a class of Bilevel Optimization (BLO) problems, with applications in areas such as signal processing, networking and machine learning. Specifically, we develop a novel barrier-based gradient approximation algorithm that transforms the constrained BLO problem to a problem with only linear equality constraints in the LL task. For the reformulated problem, we compute the implicit gradient and develop a gradient-based scheme, involving only a single gradient descent step and the (approximate) solution of the linearly constrained strongly convex LL task at each iteration. We establish, under certain assumptions, the non-asymptotic convergence guarantees of the proposed method to stationary points. Finally, we perform a number of experiments that show the potential of the proposed algorithm.

Index Terms— constrained bilevel optimization, implicit gradient, barrier function

1. INTRODUCTION

Bilevel Optimization (BLO) [1, 2] refers to a class of hierarchical problems with two levels of hierarchy, the lower (LL) and the upper-level (UL) problems (or tasks). In BLO, the solutions to the UL problem depend on the optimizer of the LL problem. BLO problems naturally arise in many applications, in areas such as economics [3], network optimization [4], game theory [5], and power systems [6]. Recently, BLO has garnered significant interest in the machine learning (ML) community because of its application in solving ML problems such as hyperparameter optimization [7], meta-learning [8], reinforcement learning [9], and adversarial learning [10].

In this work, we focus on a specific class of BLO problems where the LL task has general inequality and linear equality constraints:

$$\min_{\mathbf{x} \in \mathbb{R}^n} G(\mathbf{x}) := f(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) \quad (1)$$

$$\text{s.t. } \mathbf{y}^*(\mathbf{x}) \in \arg \min_{\mathbf{y} \in \mathbb{R}^m} g(\mathbf{x}, \mathbf{y}) \quad (2)$$

$$\text{s.t. } h_i(\mathbf{x}, \mathbf{y}) \leq 0, A\mathbf{y} = \mathbf{b}, \quad i = 1, \dots, K.$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is (possibly) a non-convex function; $g : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is strongly-convex in $\mathbf{y}$; $h_i : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is convex in $\mathbf{y}$, $\forall i \in [K]$; and $A \in \mathbb{R}^{K \times n}$ has full row rank and $\mathbf{b} \in \mathbb{R}^K$. We assume that f is one time and g, h are two times continuously differentiable functions. Note that the strong convexity of the LL objective $g(\mathbf{x}, \mathbf{y})$ ensures the uniqueness of the solution to the LL problem (for every $\mathbf{x}$), which implies that the implicit function $G(\mathbf{x})$ is well-defined.

Many applications of practical interest naturally follow the BLO formulation in (1) (see Section 1.1 for discussion and examples), therefore, it is important to develop efficient algorithms for solving this

class of problems. However, the presence of constraints in the LL task makes the implicit objective $G(\mathbf{x})$ non-differentiable, and as a result the problem difficult to tackle. To deal with this class of problems, in this work we propose to transform the constrained BLO problem (1) to a problem with only linear equality constraints in the LL problem. To accomplish this, we utilize the log-barrier function reformulation to integrate the LL inequality constraints into the LL objective. This allows us to derive a closed-form expression for the implicit gradient, and develop the Implicit Gradient with Barrier Approximation (IGBA) algorithm, a gradient-based scheme to solve the challenging constrained BLO problems of the form (1).

Related Works. There is a growing interest recently to deal with BLO problems using (implicit) gradient-based methods [2], due to their suitability in solving ML problems. However, most of implicit gradient methods are developed for problems where the LL problem is strongly-convex and unconstrained, in which the implicit gradient $\nabla G(\mathbf{x})$ can be easily derived [9, 11, 12]. On the contrary, when the LL problem has constraints, $G(\cdot)$ may no longer be differentiable. Nonetheless, there are certain special cases where one can establish the differentiability of the implicit function. For example, this is the case when the LL problem is strongly convex and has simple (i.e., independent of the UL variable $\mathbf{x}$) linear equality $A\mathbf{y} = \mathbf{b}$ [13, sec. 4.1] or inequality constraints $A\mathbf{y} \leq \mathbf{b}$ [14]. Because of the limited applicability of implicit gradient methods on constrained BLO problems, many works follow the value function approach [15–18]. Using this approach the BLO task is transformed to a single-level problem with the substitution of the LL problem with the constraint $g(\mathbf{x}, \mathbf{y}) \leq V(\mathbf{x}) = \min_{\mathbf{y} \in \mathbb{R}^m} g(\mathbf{x}, \mathbf{y})$. However, the constraint $g(\mathbf{x}, \mathbf{y}) - V(\mathbf{x})$ is in general non-differentiable and non-convex, and as a result certain modifications are typically required to make this approach usable. The resulting constrained minimization problem can then be solved using known methods such as primal-dual [17] or penalty [15, 16] methods, which can potentially be complicated.

An alternative approach that avoids the use of value function, is to directly incorporate the constraints into the LL objective using penalty/barrier functions. The works [19, 20] adopt this approach, however, these works are concerned primarily with the asymptotic properties of the solutions of the reformulated problem. Also, these works only provide simple numerical experiments, therefore the practical behavior of such methods remains unknown. In contrast, we consider a general BLO formulation (1), where in addition to general inequality constraints the LL task includes linear equality constraints that cannot be handled within the standard barrier penalization framework [19]. Importantly, we develop an implicit gradient algorithm with non-asymptotic convergence guarantees.

Contributions. In this work, we study BLO problems of form (1), where the LL task involves problems with both general inequality and linear equality constraints. These problems are challenging

because the implicit function is in general non-differentiable, and therefore computing the implicit gradient is not possible. To address this, we develop a reformulation-based approach. Our contributions are the following: 1) We provide a barrier-based penalty reformulation of problem (1), in which the implicit gradient can be computed in closed-form; 2) We develop a simple implicit gradient method, IGBA, involving only a single gradient descent step and the solution of a constrained strongly convex optimization task at each iteration; 3) We provide non-asymptotic convergence guarantees for the proposed IGBA algorithm to stationary solutions of the implicit function. To the best of our knowledge, this is the first work to provide non-asymptotic guarantees for a barrier-based BLO reformulation.

1.1. Motivating Applications

In this section, we discuss some practical applications where problems of form (1) naturally arise.

Toll Setting. Consider a transportation network graph $G = (V, E)$ and a set of weights $\{w_e\}_{e \in E}$ which capture the cost of sending traffic through each edge (e.g., value of gas). In the *toll setting* problem [21, 22], the goal is to select the toll configuration (e.g., price, position) in the network in a way such that the profit (of the toll company) is maximized. This problem is a BLO task, where in the UL problem the profit maximizing toll configuration is selected, while in the LL problem the cheapest possible way to route traffic is computed given a toll configuration.

$$\begin{aligned} \min_{\substack{\mathbf{x} \in \mathbb{R}^{|E|} \\ \mathbf{x} \geq \mathbf{0}, \mathbf{1}^T \mathbf{x} = c}} \quad & \left\{ - \sum_{(i,j) \in E} (w_{ij} + x_{ij}) y_{ij}^*(\mathbf{x}) \right\} \\ \text{s.t. } \mathbf{y}^*(\mathbf{x}) \in \quad & \begin{cases} \arg \min_{\mathbf{y} \in \mathbb{R}^{|E|}} \sum_{(i,j) \in E} (w_{ij} + x_{ij}) y_{ij} + \frac{\gamma}{2} \|\mathbf{y}\|^2 \\ \mathbf{0} \leq \mathbf{y} \leq \mathbf{b}_1, \sum_{(s,i) \in E} y_{si} = \sum_{(j,t) \in E} y_{jt} = \mathbf{b}_2. \\ \text{s.t. } \sum_{(i,j) \in E} y_{ij} - \sum_{(j,k) \in E} y_{jk} = 0, \forall j \in V \setminus \{s, t\}. \end{cases} \end{aligned}$$

The UL variable $\mathbf{x} = (x_{ij})_{(i,j) \in E}$ describes the toll amount added to every edge, while $\mathbf{c}$ is the total available budget that can be used for that purpose. Also, the LL variable $\mathbf{y} = (y_{ij})_{(i,j) \in E}$ represents the amount of traffic directed through each edge. The first LL constraint results from the fact that the traffic capacity of an edge is limited, the second constraint fixes the amount ($\mathbf{b}_2$) of traffic that is transferred from the origin (s) to the destination (t), while the last constraint ensures that the traffic that enters a node is equal to the traffic that exits. Finally, a regularizer $\frac{\gamma}{2} \|\mathbf{y}\|^2$ is added to make the LL problem strongly convex.

Jammer Attack on Power Control Problem. Consider a power control [23] problem with K users (i.e., transmitter-receiver pairs) that communicate over n channels, and a jammer. The users' goal is to allocate power across the n channels such that their joint utility (which captures the overall transmission rate) is maximized, while the jammer aims to maximize its own utility, design to harm the users' performance. This can be formulated as the following BLO problem

$$\begin{aligned} \min_{\substack{\mathbf{x} \geq \mathbf{0}, \mathbf{1}^T \mathbf{x} \leq \bar{\mathbf{x}}}} \quad & U_{\text{jam}} \left(\{R_k(\mathbf{x}, \mathbf{y})\}_{k=1}^K \right) \\ \text{s.t. } \mathbf{y}^*(\mathbf{x}) \in \quad & \begin{cases} \arg \max_{\mathbf{y} \geq \mathbf{0}, \mathbf{1}^T \mathbf{y} \leq \bar{\mathbf{y}}} U_{\text{user}} \left(\{R_k(\mathbf{x}, \mathbf{y})\}_{k=1}^K \right) \\ \text{s.t. } \mathbf{x} + \mathbf{y} \leq \mathbf{b}. \end{cases} \end{aligned} \quad (3)$$

The variables $\mathbf{x}$ and $\mathbf{y}$ describe the power allocations (across n channels), $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$ denote the total power budgets, and $U_{\text{jam}}(\cdot)$, $U_{\text{user}}(\cdot)$

are the utility functions of the users and the jammer, respectively; we use $R_k(\mathbf{x}, \mathbf{y})$ to represent the transmission rate of user k . The constraint $\mathbf{x} + \mathbf{y} \leq \mathbf{b}$ ensures that the power allocation of the users respects the upper interference limit, given the jammer's allocation. Finally, note that the bilevel formulation offers the flexibility to use different utility functions for the jammer and the users that give different priorities to fairness and system performance (e.g. see [24]).

Adversarial Learning. In an adversarial learning problem, the goal is to robustly train a model $h(\mathbf{x}, \mathbf{y})$ over a training set $\{(\mathbf{c}_i, \mathbf{d}_i)\}_{i=1}^N$; $\mathbf{x}$ is the model parameter, $\mathbf{y}$ is the input perturbation, and $\mathbf{c}_i \in \mathbb{R}^m$, $\mathbf{d}_i \in \mathbb{R}$ are the training examples. The goal of adversarial learning is to endow the model with the ability to handle adversarial examples [25]. This *adversarial training* procedure is a BLO problem stated as:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^m} \quad & \left\{ \frac{1}{2} \sum_{i=1}^N \|h(\mathbf{x}, \mathbf{c}_i + \mathbf{y}^*(\mathbf{x})) - \mathbf{d}_i\|^2 \right\} \\ \text{s.t. } \mathbf{y}^*(\mathbf{x}) \in \quad & \begin{cases} \arg \min_{\mathbf{y} \in \mathbb{R}^m} -\frac{1}{2} \sum_{i=1}^N \|h(\mathbf{x}, \mathbf{c}_i + \mathbf{y}) - \mathbf{d}_i\|^2 + \frac{\gamma}{2} \|\mathbf{y}\|^2 \\ \text{s.t. } \|\mathbf{y}\|_2 \leq b, \end{cases} \end{aligned} \quad (4)$$

where $\mathbf{y}$ is a perturbation constrained in a ball of radius b (which is a non-linear constraint) and γ is a regularization parameter; note that we use the same perturbation across all training samples.

2. BARRIER-BASED BLO REFORMULATION

We consider the following approximate reformulation of problem (1), where we use the log-barrier function $\psi_i(\mathbf{x}, \mathbf{y}) = \log[-h_i(\mathbf{x}, \mathbf{y})]$ (for the i^{th} constraint $h_i(\mathbf{x}, \mathbf{y}) \leq 0$), to integrate the LL inequality constraints into the LL objective:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^m} \quad & F(\mathbf{x}) = f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \\ \text{s.t. } \mathbf{y}_t^*(\mathbf{x}) \in \quad & \arg \min_{\mathbf{y} \in \mathbb{R}^m} \underbrace{\left\{ g(\mathbf{x}, \mathbf{y}) - \frac{1}{t} \sum_{i=1}^K \log[-h_i(\mathbf{x}, \mathbf{y})] \right\}}_{\phi_t(\mathbf{x}, \mathbf{y})} \\ \text{s.t. } \mathbf{A}\mathbf{y} &= \mathbf{b}, \end{aligned} \quad (5)$$

where $t > 0$ is a parameter that controls the accuracy of the approximation. Specifically, from the theory of interior-point methods [26], we know that $g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - g(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) \leq K/t$, where $\mathbf{y}^*(\mathbf{x})$ is the LL solution of (1). It follows directly that $\mathbf{y}_t^*(\mathbf{x}) \rightarrow \mathbf{y}^*(\mathbf{x})$ as $t \rightarrow \infty$ for any given $\mathbf{x}$.

The main benefit of reformulation (5) is that, it has only linear equality constraints, and importantly, it ensures differentiability of the implicit function $F(\mathbf{x})$ and enables the computation of the implicit gradient $\nabla F(\mathbf{x})$ in closed-form. Note that in the original form (1), the implicit function can be non-differentiable in general. We should note here that this benefit comes at the cost of solving an approximation of the original problem (1), however we have control over the approximation level through parameter t , as mentioned above. In fact, it is possible to quantify the error between the $G(\mathbf{x})$ and $F(\mathbf{x})$. To begin our analysis, let us introduce some assumptions.

Assumption 1. We assume that the following holds:

1. f is one time and g, h are two times continuously differentiable; f is lower-bounded.
2. $g(\mathbf{x}, \mathbf{y})$ is strongly convex in $\mathbf{y}$ with modulus μ , $\forall \mathbf{x} \in \mathbb{R}^n$.

3. $h(\mathbf{x}, \mathbf{y})$ is convex in $\mathbf{y}$, $\forall \mathbf{x} \in \mathbb{R}^n$.
4. For every $\mathbf{x} \in \mathbb{R}^n$ there exist $\mathbf{y} \in \mathbb{R}^m$ such that $h_i(\mathbf{x}, \mathbf{y}) < 0$, $A\mathbf{y} = \mathbf{b}$, $\forall i \in [K]$.
5. $|f(\mathbf{x}, \mathbf{y}) - f(\bar{\mathbf{x}}, \bar{\mathbf{y}})| \leq \bar{L}_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$, $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$, where $\bar{L}_f > 0$.
6. A has full row rank.

We quantify the error between the implicit functions of problem (1) and its barrier-based reformulation (5).

Lemma 1. Under Ass. 1, it holds $|G(\mathbf{x}) - F(\mathbf{x})| \leq \bar{L}_f \sqrt{\frac{2K}{\mu \cdot t}}$, $\forall \mathbf{x} \in \mathbb{R}^n$.

The next result shows the implicit gradient for $F(\mathbf{x})$, by leveraging the results from [13].

Lemma 2 (Implicit Gradient). Under Assumption 1, we have

$$\nabla F(\mathbf{x}) = \nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \quad (6)$$

$$\nabla \mathbf{y}_t^*(\mathbf{x}) = [\nabla_{yy}^2 \phi_t^*]^{-1} \left[-\nabla_{xy}^2 \phi_t^* - A^T \nabla \lambda_t^*(\mathbf{x}) \right] \quad (7)$$

$$\nabla \lambda_t^*(\mathbf{x}) = - \left[A [\nabla_{yy}^2 \phi_t^*]^{-1} A^T \right]^{-1} \left[A [\nabla_{yy}^2 \phi_t^*]^{-1} \nabla_{xy}^2 \phi_t^* \right] \quad (8)$$

$$\nabla_{yy} \phi_t(\mathbf{x}, \mathbf{y}) = \nabla_{yy} g - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_y h_i \nabla_y^T h_i}{h_i^2} + \frac{\nabla_{yy} h_i}{h_i} \right]$$

$$\nabla_{xy} \phi_t(\mathbf{x}, \mathbf{y}) = \nabla_{xy} g - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_x h_i \nabla_y^T h_i}{h_i^2} + \frac{\nabla_{xy} h_i}{h_i} \right],$$

$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in D_\phi(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^m : h_i(\mathbf{x}, \mathbf{y}) < 0, \forall i \in [K]\}$; we use the notation $g := g(\mathbf{x}, \mathbf{y})$, $h_i := h_i(\mathbf{x}, \mathbf{y})$, $\phi_t^* := \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$.

3. PROPOSED ALGORITHM AND ANALYSIS

For solving problem (5), we propose the Implicit Gradient with Barrier Approximation (IGBA) algorithm. At the start of each iteration, IGBA computes an approximate solution $\hat{\mathbf{y}}_t^*(\mathbf{x})$ of the LL problem in (5) such that $\|\hat{\mathbf{y}}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\mathbf{x})\| \leq \delta$, for a given δ . The reformulated LL problem in (5) is strongly convex, with linear equality constraints, and thus it can be easily solved to a given accuracy using known approaches, such as projected gradient descent. Then, we formulate an (approximate) implicit gradient direction $\hat{\nabla} F(\mathbf{x})$ using (6)-(8), and substituting $\mathbf{y}_t^*(\mathbf{x})$ with $\hat{\mathbf{y}}_t^*(\mathbf{x})$. Finally, a gradient descent step is performed along $\hat{\nabla} F(\mathbf{x})$. The formal description of the algorithm is given in Alg. 1. Note that the parameter t is kept fixed during the execution of the algorithm; usually a relatively large value is selected to ensure that we get an accurate approximation; see Lemma 1.

3.1. Convergence Analysis

Next, we provide convergence analysis for Algorithm 1. Before we proceed, we need the following assumptions.

Assumption 2. The following holds for every $\mathbf{x}, \bar{\mathbf{x}} \in \mathbb{R}^n, \mathbf{y}, \bar{\mathbf{y}} \in \mathbb{R}^m$, and for all $i \in [K]$,

1. $\|\nabla f(\mathbf{x}, \mathbf{y}) - \nabla f(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
2. $\|\nabla g(\mathbf{x}, \mathbf{y}) - \nabla g(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_g [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
3. $\|\nabla_{xy}^2 g(\mathbf{x}, \mathbf{y}) - \nabla_{xy}^2 g(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
4. $\|\nabla_{yy}^2 g(\mathbf{x}, \mathbf{y}) - \nabla_{yy}^2 g(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$

Algorithm 1 Implicit Gradient with Barrier Approximation (IGBA)

- 1: **Input:** $\mathbf{x}^0, \alpha, \delta, T, t$
- 2: **for** $r = 0, 1, \dots, T - 1$ **do**
- 3: Find a δ -approximate solution of the LL problem in (5)
$$\hat{\mathbf{y}}_t^*(\mathbf{x}^r) \approx \arg \min_{\substack{\mathbf{y} \in \mathbb{R}^m \\ A\mathbf{y} = \mathbf{b}}} g(\mathbf{x}^r, \mathbf{y}) - \frac{1}{t} \sum_{i=1}^K \log [-h_i(\mathbf{x}^r, \mathbf{y})]$$
- 4: Compute $\hat{\nabla} \mathbf{y}_t^*(\mathbf{x}^r)$ using eq. (6), (7), (8), where we substitute exact solution $\mathbf{y}_t^*(\mathbf{x}^r)$ with the approximate one $\hat{\mathbf{y}}_t^*(\mathbf{x}^r)$
- 5: $\hat{\nabla} F(\mathbf{x}^r) \leftarrow \nabla_x f(\mathbf{x}^r, \hat{\mathbf{y}}_t^*(\mathbf{x}^r)) + \hat{\nabla} \mathbf{y}_t^*(\mathbf{x}^r) \nabla_y f(\mathbf{x}^r, \hat{\mathbf{y}}_t^*(\mathbf{x}^r))$
- 6: Gradient Descent: $\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r - \alpha \hat{\nabla} F(\mathbf{x}^r)$
- 7: **end for**

5. $\|\nabla_{xy}^2 g(\mathbf{x}, \mathbf{y})\| \leq \bar{L}_{g_{xy}}$
6. $\|\nabla h_i(\mathbf{x}, \mathbf{y})\| \leq \bar{L}_h$
7. $\|\nabla h_i(\mathbf{x}, \mathbf{y}) - \nabla h_i(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_h [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
8. $\|\nabla_{xy}^2 h_i(\mathbf{x}, \mathbf{y}) - \nabla_{xy}^2 h_i(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_{h_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
9. $\|\nabla_{yy}^2 h_i(\mathbf{x}, \mathbf{y}) - \nabla_{yy}^2 h_i(\bar{\mathbf{x}}, \bar{\mathbf{y}})\| \leq L_{h_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y} - \bar{\mathbf{y}}\|]$
10. $\|\nabla_{xy}^2 h_i(\mathbf{x}, \mathbf{y})\| \leq \bar{L}_{h_{xy}}$
11. $\|A\| \leq L_A$

where $L_f, L_g, L_{g_{xy}}, L_{g_{yy}}, \bar{L}_{g_{xy}}, \bar{L}_h, L_h, L_{h_{xy}}, L_{h_{yy}}, \bar{L}_{h_{xy}}, L_A$ are positive constants.

The assumptions about functions f and g are conventional for gradient methods in the BLO literature, e.g., [9, 11, 27]. Similarly, the assumptions on h_i are also reasonable. In particular, if h_i 's are linear constraints in the form $h_i(\mathbf{x}, \mathbf{y}) = \mathbf{a}_i^T \mathbf{x} + \mathbf{b}_i^T \mathbf{y} + c_i \leq 0$, then all the assumptions are satisfied.

For conducting a non-asymptotic analysis of IGBA, we need one more assumption. First note that for any given $\mathbf{x}$ we know from the theory of interior-point methods that the optimal solution $\mathbf{y}_t^*(\mathbf{x})$ is strictly feasible (e.g., see [26, sec. 11.2.2]), i.e., $\exists c_{t,i}(\mathbf{x}) > 0$ s.t. $h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) < -c_{t,i}(\mathbf{x}) < 0, \forall i \in [K]$. Then, the final assumption ensures that there exists a universal (over $\mathbf{x}$ and i) bound c_t such that $-c_{t,i}(\mathbf{x}) \leq -c_t < 0, \forall \mathbf{x} \in \mathbb{R}^n, i \in [K]$.

Assumption 3. For any given $t > 0$, there exist constants $c_t, d_t > 0$ such that the following holds:

$$-d_t \leq h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \leq -c_t < 0, \quad \forall \mathbf{x} \in \mathbb{R}^n, \forall i \in [K].$$

Note that we do not have a precise expression for c_t , however since $h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \rightarrow 0$ as $t \rightarrow \infty$, we can infer that $c_t \rightarrow 0$. Therefore, it is possible to make the above assumption more precise by considering, for instance $c_t \sim 1/\sqrt{t}$, and develop a more refined analysis (see the remark after Theorem 1). Next, using the above assumptions, we show that the implicit function $F(\mathbf{x})$ is Lipschitz smooth, a property that is crucial for showing convergence.

Lemma 3. Under Ass. 1 – 3, and for all $\mathbf{x}, \bar{\mathbf{x}} \in \mathbb{R}^n$, we have

$$\|\nabla F(\mathbf{x}) - \nabla F(\bar{\mathbf{x}})\| \leq L_{F,t} \|\mathbf{x} - \bar{\mathbf{x}}\|, \quad (9)$$

where $L_{F,t} > 0$ is the Lipschitz constant that depends on K, t and the parameters of Assumptions 1, 2, 3.

Note that in Algorithm 1, we only compute an approximate implicit gradient $\hat{\nabla} F(\mathbf{x})$, which results from substituting $\mathbf{y}_t^*(\mathbf{x})$ in eq. (6)-(8) with the approximate solution $\hat{\mathbf{y}}_t^*(\mathbf{x})$. Below we provide the error bound between the exact and the approximate implicit gradient.

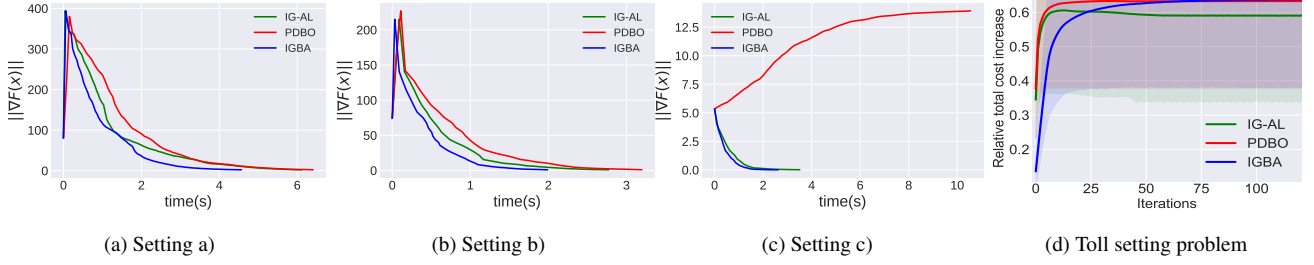


Fig. 1: Plots for the numerical (1a,1b,1c) and the toll setting problem (1d) experiments. IGBA is the proposed algorithm.

Lemma 4. Suppose that Ass. 1-3 holds, and that Ass. 3 is also valid for the approximate solution $\hat{\mathbf{y}}_t^*(\mathbf{x})$. Then $\forall \mathbf{x} \in \mathbb{R}^n$ it holds that

$$\|\nabla F(\mathbf{x}) - \hat{\nabla} F(\mathbf{x})\| \leq \hat{L}_{F,t} \cdot \delta \quad (10)$$

where δ is defined in Algorithm 1 as $\|\hat{\mathbf{y}}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\mathbf{x})\| \leq \delta, \forall \mathbf{x} \in \mathbb{R}^n$ and $\hat{L}_{F,t} > 0$ is a constant that depends on K, t and the parameters of Assumptions 1, 2, 3.

Finally, we present the main result establishing the convergence of the IGBA algorithm to a stationary point of the implicit function $F(\mathbf{x})$ with a $1/T$ rate up to an additive error term that depends on the LL solution accuracy δ .

Theorem 1 (Convergence of IGBA). Suppose that Assumptions 1-3 hold and $\alpha = \frac{5}{4L_{F,t}}$. Then, after T iterations of the IGBA, we obtain

$$\frac{1}{T} \sum_{r=0}^{T-1} \|\nabla F(\mathbf{x}^r)\|^2 \leq \frac{16}{5T} (F(\mathbf{x}^0) - \underline{F}) L_{F,t} + 3\hat{L}_{F,t} \cdot \delta^2, \quad (11)$$

where $\underline{F}$ is a lower bound on F , and the constants $L_{F,t}, \hat{L}_{F,t}$ are defined in Lemmas 3 and 4, respectively.

Proof. Due to space limitations we omit the proofs of Lemmas 1-4 and Theorem 1. The readers can find the proofs in [28]. $\square$

Remark. Note that the constants $L_{F,t}, \hat{L}_{F,t}$ are functions of (among other things) t and c_t ; we also know $c_t \rightarrow 0$ as $t \rightarrow \infty$ since $h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \rightarrow 0$. Then, if we assume that $c_t = q_1/\sqrt{t}$, for some constant q_1 , it can be shown that $L_F \leq q_2 \cdot t$ and $\hat{L}_{F,t} \leq q_3 \cdot t$ (where q_2, q_3 are constants). Combining this observation with (11) we get the following convergence guarantee: $\frac{1}{T} \sum_{r=0}^{T-1} \|\nabla F(\mathbf{x}^r)\|^2 \leq \frac{16q_2(F(\mathbf{x}^0) - \underline{F})}{5} \frac{t}{T} + 3q_3 \cdot t \cdot \delta^2$. This result captures the trade-off between the approximation error of Lemma 1 and the convergence performance of IGBA. Specifically, this result implies that the larger the value of parameter t (implying small approximation error; see Lemma 1), the higher the number of iterations T and the lower the LL approximation error δ , needed to achieve ϵ -stationarity, for a given $\epsilon > 0$.

4. EXPERIMENTS

Numerical Examples. We consider the following three settings: a) $f(\mathbf{x}, \mathbf{y}) = \frac{1}{8}\|\mathbf{x}\|^2 + 10\mathbf{x}^T\mathbf{y} + \frac{1}{8}\|\mathbf{y}\|^2, g(\mathbf{x}, \mathbf{y}) = \frac{1}{2}\mathbf{x}^T\mathbf{y} + \frac{1}{4}\|\mathbf{y}\|^2$; b) $f(\mathbf{x}, \mathbf{y}) = \frac{1}{4}\|\mathbf{x}\|^2 + 5\mathbf{x}^T\mathbf{y} - \frac{1}{4}\|\mathbf{y}\|^2 + \mathbf{1}^T\mathbf{x} + \mathbf{1}^T\mathbf{y} + 1, g(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y} + \frac{1}{2}\|\mathbf{y}\|^2 + \mathbf{1}^T\mathbf{x} + \mathbf{1}^T\mathbf{y}$; c) $f(\mathbf{x}, \mathbf{y}) = -\frac{1}{4}\|\mathbf{x}\|^2 - \frac{1}{2}\mathbf{x}^T\mathbf{y}, g(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y} + \frac{1}{2}\|\mathbf{y}\|^2 + \mathbf{1}^T\mathbf{x} + \mathbf{1}^T\mathbf{y}$. Also, the LL constraints are of the form $|y_i| \leq 5, \forall i$, that is we have only inequality constraints, and $n = m = 50$. We compare the proposed IGBA method

with the following baselines: 1) IG-AL [14] (an implicit gradient method for linearly constrained problems), 2) PDBO [17] (a primal-dual value-function-based method). In setting a) and b) we run all algorithms for 100 and 50 iterations, respectively; in setting c) we execute IG-AL for 20 iterations, PDBO for 80 iterations, and IGBA for 60 iterations. Further, we set $t = 10^3, 10^6, 10^8$ for the experiments of setting a), b) and c), respectively. Finally, for the parameter values we have the following: in settings a) and b) the stepsizes for all three algorithms are set to 0.2; in setting c) the stepsizes are set to 0.5, 0.1 and 0.2 for IG-AL, PDBO, IGBA, respectively.

We plot in Figs. 1a, 1b, 1c the norm of the implicit gradient $\|\nabla F(\mathbf{x})\|$ as a function of the elapsed time, for settings a), b), and c) respectively. In Figs. 1a, 1b we observe that all algorithms converge to stationary points, however it appears that the proposed IGBA method converges faster compared to the other algorithms. In the (more difficult, due to the non-convexity of f in $\mathbf{x}$) problem in Fig. 1c the PDBO method does not converge to a stationary solution. To summarize, the proposed IGBA algorithm performs well over all the cases we tested. More precisely, IGBA converges faster compared to the other methods in all experiments (Figs. 1a, 1b, 1c), and converges even in the case of a difficult problem (Fig. 1c) with a non-convex UL objective where PDBO fails.

Toll Setting. In the toll setting problem we generate a random graph with $n = 12$ nodes using the Erdos-Renyi model (the probability that an edge appears is $p = 0.8$). Further, the weights, capacity and total amount of traffic (to be routed through the network) are generated in the following manner: the weights and the capacity $\mathbf{b}_1$ of each edge are sampled from uniform distributions in the intervals $[5, 30]$ and $[100, 1000]$, respectively, while the amount of traffic $\mathbf{b}_2$ is selected as 75% of the sum of capacities that exit the source. We compare IGBA with PDBO [17] and IG-AL [14]; all algorithms are executed for $T = 125$ iterations and the stepsizes are set to 10^{-2} for IG-AL and PDBO, while for IGBA we have $\alpha = 2 \cdot 10^{-3}$. The algorithms' performance is evaluated in the following manner: 1) We run the algorithms and compute the toll allocation $\mathbf{x}$ for different edges; 2) We solve the minimum cost flow problem on the new graph (with the addition of tolls) and obtain the "cheapest" traffic allocation $\mathbf{y}$; 3) We compute the relative total cost increase as follows $(\text{cost}_{\text{after}} - \text{cost}_{\text{before}}) / \text{cost}_{\text{before}}$, where $\text{cost}_{\text{before}}$, $\text{cost}_{\text{after}}$ are the costs of the "cheapest" traffic allocation before and after the addition of tolls, respectively.

We execute the same experiment, generating a new graph each time, 20 times. In figure 1d we plot the (average) relative total cost increase with respect to the number of iterations, for the case where the total toll budget is set to $c = 100$. The range of values obtained in the 20 runs is displayed using the shaded area around the average line. Finally, notice that IGBA performs better than IG-AL, while it exhibits almost the same performance as PDBO.

5. REFERENCES

- [1] Benoît Colson, Patrice Marcotte, and Gilles Savard, “Bilevel programming: A survey,” *4or*, vol. 3, no. 2, pp. 87–107, 2005.
- [2] Risheng Liu, Jiabin Gao, Jin Zhang, Deyu Meng, and Zhouchen Lin, “Investigating bi-level optimization for learning and vision from a unified perspective: A survey and beyond,” *arXiv*, 2021.
- [3] Mark Cecchini, Joseph Ecker, Michael Kupferschmid, and Robert Leitch, “Solving nonlinear principal-agent problems using bilevel programming,” *European Journal of Operational Research*, vol. 230, no. 2, pp. 364–373, 2013.
- [4] Athanasios Migdalas, “Bilevel programming in traffic planning: Models, methods and challenge,” *Journal of global optimization*, vol. 7, no. 4, pp. 381–405, 1995.
- [5] Heinrich Von Stackelberg, *The theory of the market economy*, Oxford University Press, 1952.
- [6] Amin Abedi, Mohammad Reza Hesamzadeh, and Franco Romero, “An acopf-based bilevel optimization approach for vulnerability assessment of a power system,” *International Journal of Electrical Power & Energy Systems*, vol. 125, pp. 106455, 2021.
- [7] Ankur Sinha, Tanmay Khandait, and Raja Mohanty, “A gradient-based bilevel optimization approach for tuning hyperparameters in machine learning,” *arXiv preprint arXiv:2007.11022*, 2020.
- [8] Aravind Rajeswaran, Chelsea Finn, Sham M Kakade, and Sergey Levine, “Meta-learning with implicit gradients,” in *Advances in Neural Information Processing Systems*. 2019, vol. 32, Curran Associates, Inc.
- [9] Mingyi Hong, Hoi-To Wai, Zhaoran Wang, and Zhuoran Yang, “A two-timescale framework for bilevel optimization: Complexity analysis and application to actor-critic,” *arXiv preprint arXiv:2007.05170*, 2020.
- [10] Yihua Zhang, Guanhua Zhang, Prashant Khanduri, Mingyi Hong, Shiyu Chang, and Sijia Liu, “Revisiting and advancing fast adversarial training through the lens of bi-level optimization,” *arXiv: 2112.12376*, 2021.
- [11] Saeed Ghadimi and Mengdi Wang, “Approximation methods for bilevel programming,” *arXiv preprint arXiv:1802.02246*, 2018.
- [12] Prashant Khanduri, Siliang Zeng, Mingyi Hong, Hoi-To Wai, Zhaoran Wang, and Zhuoran Yang, “A momentum-assisted single-timescale stochastic approximation algorithm for bilevel optimization,” 2021.
- [13] Stephen Gould, Basura Fernando, Anoop Cherian, Peter Anderson, Rodrigo Santa Cruz, and Edison Guo, “On differentiating parameterized argmin and argmax problems with application to bi-level optimization,” 2016.
- [14] Ioannis Tsaknakis, Prashant Khanduri, and Mingyi Hong, “An implicit gradient-type method for linearly constrained bilevel problems,” in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 5438–5442.
- [15] Gui-Hua Lin, Mengwei Xu, and Jane J. Ye, “On solving simple bilevel programs with a nonconvex lower level program,” *Math. Program.*, vol. 144, no. 1–2, pp. 277–305, Apr. 2014.
- [16] Risheng Liu, Xuan Liu, Xiaoming Yuan, Shangzhi Zeng, and Jin Zhang, “A value-function-based interior-point method for non-convex bi-level optimization,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 6882–6892.
- [17] Daouda Sow, Kaiyi Ji, Ziwei Guan, and Yingbin Liang, “A constrained optimization approach to bilevel optimization with multiple inner minima,” *arXiv preprint arXiv:2203.01123*, 2022.
- [18] Risheng Liu, Xuan Liu, Shangzhi Zeng, Jin Zhang, and Yixuan Zhang, “Value-function-based sequential minimization for bi-level optimization,” *arXiv preprint arXiv:2110.04974*, 2021.
- [19] K. Shimizu and E. Aiyoshi, “A new computational method for stackelberg and min-max problems by use of a penalty method,” *IEEE Transactions on Automatic Control*, vol. 26, no. 2, pp. 460–466, 1981.
- [20] Yo Ishizuka and Eitaro Aiyoshi, “Double penalty method for bilevel optimization problems,” *Annals of Operations Research*, vol. 34, no. 1, pp. 73–88, 1992.
- [21] Mohamed Didi-Biha, Patrice Marcotte, and Gilles Savard, *Path-based formulations of a bilevel toll setting problem*, pp. 29–50, Springer US, Boston, MA, 2006.
- [22] Dr. Vyacheslav Kalashnikov, José-Fernando Camacho-Vallejo, Ronald Askin, and Nataliya Kalashnykova, “Comparison of algorithms for solving a bi-level toll setting problem,” *International journal of innovative computing, information & control: IJICIC*, vol. 6, pp. 3529–3549, 08 2010.
- [23] Ramy H. Gohary, Yao Huang, Zhi-Quan Luo, and Jong-Shi Pang, “A generalized iterative water-filling algorithm for distributed power control in the presence of a jammer,” *IEEE Transactions on Signal Processing*, vol. 57, no. 7, pp. 2660–2674, 2009.
- [24] Mingyi Hong and Zhi-Quan Luo, “Signal processing and optimal resource allocation for the interference channel,” in *Academic Press Library in Signal Processing*, vol. 2, pp. 409–469. Elsevier, 2014.
- [25] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy, “Explaining and harnessing adversarial examples,” *arXiv preprint arXiv:1412.6572*, 2014.
- [26] S. Boyd and L. Vandenberghe, *Convex optimization*, Cambridge university press, 2004.
- [27] Prashant Khanduri, Siliang Zeng, Mingyi Hong, Hoi-To Wai, Zhaoran Wang, and Zhuoran Yang, “A near-optimal algorithm for stochastic bilevel optimization via double-momentum,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 30271–30283, 2021.
- [28] Ioannis Tsaknakis, Prashant Khanduri, and Mingyi Hong, “An implicit gradient method for constrained bilevel problems using barrier approximation: supplementary material,” 2022, https://people.ece.umn.edu/~mhong/barrier_blo_full.pdf.

A. APPENDIX

A.1. Proof of Lemma 1

Proof of Lemma 1. To begin with, we know that the LL objective $g(\mathbf{x}, \mathbf{y})$ is strongly convex in $\mathbf{y}$. Therefore,

$$g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \geq g(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) + \langle \nabla_y g(\mathbf{x}, \mathbf{y}^*(\mathbf{x})), \mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x}) \rangle + \frac{\mu}{2} \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x})\|^2. \quad (12)$$

The fact that $\mathbf{y}^*(\mathbf{x})$ is the minimum of the LL problem in (1) implies that $\langle \nabla_y g(\mathbf{x}, \mathbf{y}^*(\mathbf{x})), \mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x}) \rangle \geq 0$. Combining this with equation (12) results to

$$\|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x})\|^2 \leq \frac{2}{\mu} (g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - g(\mathbf{x}, \mathbf{y}^*(\mathbf{x}))). \quad (13)$$

Moreover, from interior-point methods theory [26, pg. 566], we know that $g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - g(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) \leq K/t$. Plugging the last inequality into (13) gives

$$\|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x})\| \leq \sqrt{\frac{2K}{\mu t}}. \quad (14)$$

Next, using Assumption 1.5 and the implicit functions of the problems (1) and (5) we get

$$\begin{aligned} |G(\mathbf{x}) - F(\mathbf{x})| &= |f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - f(\mathbf{x}, \mathbf{y}^*(\mathbf{x}))| \\ &\leq \bar{L}_f \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}^*(\mathbf{x})\| \\ &\leq \bar{L}_f \sqrt{\frac{2K}{\mu t}}, \end{aligned}$$

where in the last inequality we used equation (14). The proof is complete. $\square$

A.2. Proof of Lemma 2.

Proof of Lemma 2. The gradient of the implicit function $F(\mathbf{x})$ of problem (5) follows from the application of the chain rule

$$\nabla F(\mathbf{x}) = \nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$$

where

$$\begin{aligned} \nabla \mathbf{y}_t^*(\mathbf{x}) &= [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \\ &\quad [-\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - A^T \nabla \lambda_t^*(\mathbf{x})] \\ \nabla \lambda_t^*(\mathbf{x}) &= -[A[\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} A^T]^{-1} \\ &\quad [A[\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))] \end{aligned}$$

The last two expressions describe the Jacobian of the mapping $\mathbf{y}^*(\mathbf{x})$ when the LL problems has linear equality constraints of the form $A\mathbf{y} = \mathbf{b}$ (see [13, Lemma 4.2]).

Moreover, for the gradients and Hessians of the reformulated LL objective we have

$$\nabla_y \phi_t(\mathbf{x}, \mathbf{y}) = \nabla_y g(\mathbf{x}, \mathbf{y}) - \frac{1}{t} \sum_{i=1}^K \frac{\nabla_y h_i(\mathbf{x}, \mathbf{y})}{h_i(\mathbf{x}, \mathbf{y})}$$

and

$$\begin{aligned} \nabla_{yy} \phi_t(\mathbf{x}, \mathbf{y}) &= \nabla_{yy} g(\mathbf{x}, \mathbf{y}) - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_y h_i(\mathbf{x}, \mathbf{y}) \nabla_y^T h_i(\mathbf{x}, \mathbf{y})}{h_i^2(\mathbf{x}, \mathbf{y})} \right. \\ &\quad \left. + \frac{\nabla_{yy} h_i(\mathbf{x}, \mathbf{y})}{h_i(\mathbf{x}, \mathbf{y})} \right] \\ \nabla_{xy} \phi_t(\mathbf{x}, \mathbf{y}) &= \nabla_{xy} g(\mathbf{x}, \mathbf{y}) - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_x h_i(\mathbf{x}, \mathbf{y}) \nabla_y^T h_i(\mathbf{x}, \mathbf{y})}{h_i^2(\mathbf{x}, \mathbf{y})} \right. \\ &\quad \left. + \frac{\nabla_{xy} h_i(\mathbf{x}, \mathbf{y})}{h_i(\mathbf{x}, \mathbf{y})} \right]. \end{aligned}$$

The proof is complete. $\square$

A.3. Proof of Lemma 3

In order to prove Lemma 3,4 and Theorem 1 we need some preliminary derivations.

Lemma 5. *The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$.*

1. $\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \succeq \mu I$
2. $\|[\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1}\| \leq \frac{1}{\mu}$
3. $\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \leq \bar{L}_{\phi_{yy}}$
4. $\|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \leq \bar{L}_{\phi_{xy}}$
5. $\left\| \left[A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} A^T \right]^{-1} \right\| \leq \bar{L}_{A, \phi_t^{-1}}$

Proof. In the following proofs we use the notation $\mathbf{y}_t^* := \mathbf{y}_t^*(\mathbf{x})$.

1. The log-barrier term $\log[-h_i(\mathbf{x}, \mathbf{y})]$ is strictly concave and therefore Ass. 1.2 implies that

$$\nabla_{yy}^2 g(\mathbf{x}, \mathbf{y}_t^*) \succeq \mu I \implies \nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) \succeq \mu I.$$

2. Using the above result we can easily see the following

$$\begin{aligned} \nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) \succeq \mu I &\implies [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*)]^{-1} \preceq \frac{1}{\mu} I \\ &\implies \|[\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*)]^{-1}\| \leq \frac{1}{\mu}. \end{aligned}$$

3. We use the following notation: $g^* := g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$, $h_i^* := h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$. Then, we have

$$\begin{aligned} &\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)\| \\ &= \left\| \nabla_{yy} g^* - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_y h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} + \frac{\nabla_{yy} h_i^*}{h_i^*} \right] \right\| \\ &\leq \|\nabla_{yy} g^*\| + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \|\nabla_y h_i^*\|^2 + \left| \frac{1}{h_i^*} \right| \|\nabla_{yy} h_i^*\| \right] \\ &\leq L_g + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{c_t^2} \bar{L}_h^2 + \frac{1}{c_t} L_h \right] \\ &\leq L_g + \frac{K}{t} \left[\frac{1}{c_t^2} \bar{L}_h^2 + \frac{1}{c_t} L_h \right] := \bar{L}_{\phi_{yy}}. \end{aligned}$$

4. For the Hessian of $\phi_t(\mathbf{x}, \mathbf{y})$ w.r.t. $\mathbf{xy}$ at $(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$ we have

$$\begin{aligned} & \|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)\| \\ &= \left\| \nabla_{xy} g^* - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_x h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} + \frac{\nabla_{xy} h_i^*}{h_i^*} \right] \right\| \\ &\leq \|\nabla_{xy} g^*\| + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \|\nabla_x h_i^*\| \|\nabla_y h_i^*\| + \left| \frac{1}{h_i^*} \right| \|\nabla_{xy} h_i^*\| \right] \\ &\leq \bar{L}_{g_{xy}} + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{c_t^2} \bar{L}_h^2 + \frac{1}{c_t} \bar{L}_{h_{xy}} \right] \\ &\leq \bar{L}_{g_{xy}} + \frac{K}{t} \left[\frac{1}{c_t^2} \bar{L}_h^2 + \frac{1}{c_t} \bar{L}_{h_{xy}} \right] := \bar{L}_{\phi_{xy}}. \end{aligned}$$

5. We already established above (in 1. and 3.) that the following holds for any $\mathbf{x} \in \mathbb{R}^n$

$$\begin{aligned} & \bar{L}_{\phi_{yy}} I \succeq \nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*) \succeq \mu I \succ 0 \\ \implies 0 & \prec \frac{1}{\bar{L}_{\phi_{yy}}} I \preceq [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)]^{-1} \preceq \frac{1}{\mu} I. \end{aligned}$$

Also, note that

$$\|A^T \mathbf{z}\|^2 = \mathbf{z}^T A A^T \mathbf{z} \geq \lambda_{\min}(A A^T) \|\mathbf{z}\|^2, \forall \mathbf{z} \in \mathbb{R}^n.$$

Using the result above we obtain

$$\begin{aligned} & \mathbf{z}^T A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)]^{-1} A^T \mathbf{z} \\ & \geq \frac{1}{\bar{L}_{\phi_{yy}}} \lambda_{\min}(A A^T) \|\mathbf{z}\|^2 > 0, \forall \mathbf{z} \in \mathbb{R}^m \setminus \{0\}. \end{aligned}$$

The second inequality follows from the full-rankness assumption of A (Ass. 1.6) which further implies that $\lambda_{\min}(A A^T) > 0$. Then, because the above inequality holds for every $\mathbf{z} \in \mathbb{R}^m \setminus \{0\}$, we get that

$$\begin{aligned} & A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)]^{-1} A^T \succeq \frac{\lambda_{\min}(A A^T)}{\bar{L}_{\phi_{yy}}} I \succ 0 \\ & \left[A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)]^{-1} A^T \right]^{-1} \preceq \frac{\bar{L}_{\phi_{yy}}}{\lambda_{\min}(A A^T)} I \\ & \left\| \left[A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}^*)]^{-1} A^T \right]^{-1} \right\| \leq \frac{\bar{L}_{\phi_{yy}}}{\lambda_{\min}(A A^T)} := \bar{L}_{A, \phi_t^{-1}}. \end{aligned}$$

The proof is complete. $\square$

Lemma 6. The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$

1. $\|\nabla \lambda_t^*(\mathbf{x})\| \leq \bar{L}_{A, \phi_t^{-1}} L_A \frac{1}{\mu} \bar{L}_{\phi_{xy}} := \bar{L}_{\lambda^*}$
2. $\|\nabla \mathbf{y}_t^*(\mathbf{x})\| \leq \frac{1}{\mu} [\bar{L}_{\phi_{xy}} + L_A \bar{L}_{\lambda^*}] := \bar{L}_{y^*}$

Proof. 1. For the norm of the quantity $\nabla \lambda_t^*(\mathbf{x})$ we can compute the following bound

$$\begin{aligned} \|\nabla \lambda_t^*(\mathbf{x})\| &= \left\| \left[A [\nabla_{yy}^2 \phi_t^*(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} A^T \right]^{-1} \right. \\ & \quad \left. \left[A [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \right] \right\| \\ &\leq \left\| \left[A [\nabla_{yy}^2 \phi_t^*(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} A^T \right]^{-1} \right\| \\ & \quad \left\| A \left\| [\nabla_{yy}^2 \phi_t^*(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \right\| \left\| \nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \right\| \right\| \\ &\leq \bar{L}_{A, \phi_t^{-1}} L_A \frac{1}{\mu} \bar{L}_{\phi_{xy}} := \bar{L}_{\lambda^*} \end{aligned}$$

2. For the norm of the quantity $\nabla \mathbf{y}_t^*(\mathbf{x})$ we can compute the following bound

$$\begin{aligned} \|\nabla \mathbf{y}_t^*(\mathbf{x})\| &= \left\| [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \cdot \right. \\ & \quad \left. \cdot \left[-\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - A^T \nabla \lambda_t^*(\mathbf{x}) \right] \right\| \\ &\leq \left\| [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))]^{-1} \right\| \left\| \left[\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \right] + \|A\| \|\nabla \lambda_t^*(\mathbf{x})\| \right\| \\ &\leq \frac{1}{\mu} [\bar{L}_{\phi_{xy}} + L_A \bar{L}_{\lambda^*}] := \bar{L}_{y^*} \end{aligned}$$

$\square$

Lemma 7. The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$

1. $\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| \leq L_{\phi_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|]$
2. $\|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{xy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| \leq L_{\phi_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|],$

where $\mathbf{y}_t^* = \mathbf{y}_t^*(\mathbf{x})$, $\bar{\mathbf{y}}_t^* = \mathbf{y}_t^*(\bar{\mathbf{x}})$

Proof. We use the following notation: $g^* := g(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$, $h_i^* := h_i(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$ and $\bar{g}^* := g(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))$, $\bar{h}_i^* := h_i(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))$

1. To begin with, it holds that

$$\begin{aligned} & \|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\ &= \left\| \nabla_{yy} g^* - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_y h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} + \frac{\nabla_{yy} h_i^*}{h_i^*} \right] \right. \\ & \quad \left. - \nabla_{yy} \bar{g}^* + \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_y \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} + \frac{\nabla_{yy} \bar{h}_i^*}{\bar{h}_i^*} \right] \right\| \\ &\stackrel{(a)}{\leq} L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\ & \quad + \frac{1}{t} \sum_{i=1}^K \left\| \frac{\nabla_y h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} - \frac{\nabla_y \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} \right\| + \frac{1}{t} \sum_{i=1}^K \left\| \frac{\nabla_{yy} h_i^*}{h_i^*} - \frac{\nabla_{yy} \bar{h}_i^*}{\bar{h}_i^*} \right\| \\ &\stackrel{(b)}{\leq} L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\ & \quad + \frac{1}{t} \sum_{i=1}^K \left[\left\| \frac{\nabla_y h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} - \frac{\nabla_y \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} \right\| + \left\| \frac{\nabla_{yy} h_i^*}{(h_i^*)^2} - \frac{\nabla_{yy} \bar{h}_i^*}{(\bar{h}_i^*)^2} \right\| \right] \\ & \quad + \frac{1}{t} \sum_{i=1}^K \left[\left\| \frac{\nabla_{yy} h_i^*}{h_i^*} - \frac{\nabla_{yy} \bar{h}_i^*}{\bar{h}_i^*} \right\| + \left\| \frac{\nabla_{yy} h_i^*}{h_i^*} - \frac{\nabla_{yy} \bar{h}_i^*}{\bar{h}_i^*} \right\| \right], \end{aligned}$$

where in (a) we use Ass. 2.4 and the triangle inequality, and in (b)

we add and subtract terms and use the triangle inequality. Moreover,

$$\begin{aligned}
& \|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\
& \stackrel{(a)}{\leq} L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \left\| \nabla_y h_i^* \nabla_y^T h_i^* - \nabla_y h_i^* \nabla_y^{T*} \right\| \right. \\
& \left. + \left\| \nabla_y h_i^* \nabla_y^{T*} - \nabla_y^* \nabla_y^{T*} \right\| \right. \\
& \left. + \left\| \nabla_y^* \nabla_y^{T*} \right\| \left| \frac{1}{(h_i^*)^2} - \left(\frac{1}{\bar{h}_i^*} \right)^2 \right| \right] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{\bar{h}_i^*} \left\| \nabla_{yy} h_i^* - \nabla_{yy} \bar{h}_i^* \right\| + \left\| \nabla_{yy}^* \right\| \left| \frac{1}{\bar{h}_i^*} - \frac{1}{h_i^*} \right| \right] \\
& \stackrel{(b)}{\leq} L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \left(\left\| \nabla_y h_i^* \right\| \left\| \nabla_y h_i^* - \nabla_y^* \right\| \right. \right. \\
& \left. \left. + \left\| \nabla_y h_i^* - \nabla_y^* \right\| \left\| \nabla_y^* \right\| \right) \right. \\
& \left. + \left\| \nabla_y^* \nabla_y^{T*} \right\| \left| \frac{1}{(h_i^*)^2} - \left(\frac{1}{\bar{h}_i^*} \right)^2 \right| \left| h_i^* - \bar{h}_i^* \right| \right] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{\bar{h}_i^*} \left\| \nabla_{yy} h_i^* - \nabla_{yy} \bar{h}_i^* \right\| + \left\| \nabla_{yy}^* \right\| \left| \frac{\bar{h}_i^* - h_i^*}{\bar{h}_i^* h_i^*} \right| \right],
\end{aligned}$$

where in (a) we add and subtract terms and use the triangle inequality. Finally, applying Ass. 2.6, 2.7, 3 in the second/third line, Ass. 2.6, 3 in the fourth line, and Ass. 2.9, 2.7, 2.6, Ass. 3 in the fifth line, in (b) above we obtain the following.

$$\begin{aligned}
& \|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\
& \leq L_{g_{yy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{2}{c_t^2} \bar{L}_h L_h + \bar{L}_h^3 \frac{2d}{c_t^4} \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{c_t} L_{h_{yy}} + L_h \bar{L}_h \frac{1}{c_t^2} \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& \leq \left[L_{g_{yy}} + \frac{K}{t} \left(\frac{2}{c_t^2} \bar{L}_h L_h + \bar{L}_h^3 \frac{2d}{c_t^4} \right) \right. \\
& \left. + \frac{K}{t} \left(\frac{1}{c_t} L_{h_{yy}} + L_h \bar{L}_h \frac{1}{c_t^2} \right) \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|].
\end{aligned}$$

The proof is complete.

2. To begin with, it holds that

$$\begin{aligned}
& \|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{xy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\
& = \left\| \nabla_{xy} g^* - \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_x h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} + \frac{\nabla_{xy} h_i^*}{h_i^*} \right] \right. \\
& \quad \left. - \nabla_{xy} \bar{g}^* + \frac{1}{t} \sum_{i=1}^K \left[-\frac{\nabla_x \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} + \frac{\nabla_{xy} \bar{h}_i^*}{\bar{h}_i^*} \right] \right\| \\
& \stackrel{(a)}{\leq} L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left\| \frac{\nabla_x h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} - \frac{\nabla_x \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} \right\| + \frac{1}{t} \sum_{i=1}^K \left\| \frac{\nabla_{xy} h_i^*}{h_i^*} - \frac{\nabla_{xy} \bar{h}_i^*}{\bar{h}_i^*} \right\| \\
& \stackrel{(b)}{\leq} L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\left\| \frac{\nabla_x h_i^* \nabla_y^T h_i^*}{(h_i^*)^2} - \frac{\nabla_x^* \nabla_y^{T*}}{(h_i^*)^2} \right\| \right. \\
& \left. + \left\| \frac{\nabla_x^* \nabla_y^{T*}}{(h_i^*)^2} - \frac{\nabla_x \bar{h}_i^* \nabla_y^T \bar{h}_i^*}{(\bar{h}_i^*)^2} \right\| \right] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\left\| \frac{\nabla_{xy} h_i^*}{h_i^*} - \frac{\nabla_{xy} \bar{h}_i^*}{\bar{h}_i^*} \right\| + \left\| \frac{\nabla_{xy}^*}{h_i^*} - \frac{\nabla_{xy} \bar{h}_i^*}{\bar{h}_i^*} \right\| \right],
\end{aligned}$$

where in (a) we use Ass. 2.3 and the triangle inequality, and in (b) we add and subtract terms and use the triangle inequality. Moreover,

$$\begin{aligned}
& \|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{xy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\
& \stackrel{(a)}{\leq} L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \left\| \nabla_x h_i^* \nabla_y^T h_i^* - \nabla_x h_i^* \nabla_y^{T*} \right\| \right. \\
& \left. + \left\| \nabla_x h_i^* \nabla_y^{T*} - \nabla_x^* \nabla_y^{T*} \right\| \right. \\
& \left. + \left\| \nabla_x^* \nabla_y^{T*} \right\| \left| \frac{1}{(h_i^*)^2} - \left(\frac{1}{\bar{h}_i^*} \right)^2 \right| \right] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{\bar{h}_i^*} \left\| \nabla_{xy} h_i^* - \nabla_{xy} \bar{h}_i^* \right\| + \left\| \nabla_{xy}^* \right\| \left| \frac{1}{\bar{h}_i^*} - \frac{1}{h_i^*} \right| \right] \\
& \stackrel{(b)}{\leq} L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{(h_i^*)^2} \left(\left\| \nabla_x h_i^* \right\| \left\| \nabla_y h_i^* - \nabla_y^* \right\| \right. \right. \\
& \left. \left. + \left\| \nabla_x h_i^* - \nabla_x^* \right\| \left\| \nabla_y^* \right\| \right) \right. \\
& \left. + \left\| \nabla_x^* \nabla_y^{T*} \right\| \left| \frac{1}{(h_i^*)^2} - \left(\frac{1}{\bar{h}_i^*} \right)^2 \right| \left| h_i^* - \bar{h}_i^* \right| \right] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{\bar{h}_i^*} \left\| \nabla_{xy} h_i^* - \nabla_{xy} \bar{h}_i^* \right\| + \left\| \nabla_{xy}^* \right\| \left| \frac{\bar{h}_i^* - h_i^*}{\bar{h}_i^* h_i^*} \right| \right]
\end{aligned}$$

where in (b) we add and subtract terms and use the triangle inequality. Finally, applying Ass. 2.6, 2.7, 3 in the second/third line, Ass.

2.6, 3 in the fourth line, and Ass. 2.8, 2.10, 2.6, 3 in the fifth line, in (b) above we obtain the following.

$$\begin{aligned}
& \|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{xy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)\| = \\
& \leq L_{g_{xy}} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{2}{c_t^2} \bar{L}_h L_h + \bar{L}_h^3 \frac{2d}{c_t^4} \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& + \frac{1}{t} \sum_{i=1}^K \left[\frac{1}{c_t} L_{h_{xy}} + L_h \bar{L}_h \frac{1}{c_t^2} \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \\
& \leq \left[L_{g_{xy}} + \frac{K}{t} \left(\frac{2}{c_t^2} \bar{L}_h L_h + \bar{L}_h^3 \frac{2d}{c_t^4} \right) \right. \\
& \left. + \frac{K}{t} \left(\frac{1}{c_t} L_{h_{xy}} + \bar{L}_{h_{xy}} \bar{L}_h \frac{1}{c_t^2} \right) \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|].
\end{aligned}$$

The proof is complete. $\square$

Lemma 8. *The following hold for every $\mathbf{x}, \bar{\mathbf{x}} \in \mathbb{R}^n$*

1. $\|\nabla \lambda_t^*(\mathbf{x}) - \nabla \lambda_t^*(\bar{\mathbf{x}})\| \leq L_{\lambda^*} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|]$
2. $\|\nabla \mathbf{y}_t^*(\mathbf{x}) - \nabla \mathbf{y}_t^*(\bar{\mathbf{x}})\| \leq L_{y^*} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|]$

where

$$\begin{aligned}
L_{\lambda^*} &= \bar{L}_{A, \phi_t^{-1}} \frac{L_A^2 L_{\phi yy}}{\mu^2} \bar{L}_{A, \phi_t^{-1}} L_A \frac{1}{\mu} \bar{L}_{\phi xy} \\
&+ \bar{L}_{A, \phi_t^{-1}} L_A \left(\frac{L_{\phi yy}}{\mu^2} \bar{L}_{\phi xy} + \frac{1}{\mu} L_{\phi xy} \right).
\end{aligned}$$

and

$$L_{y^*} = \frac{\bar{L}_{\phi xy} L_{\phi yy}}{\mu^2} + \frac{L_{\phi xy}}{\mu} + \frac{L_{\phi yy}}{\mu^2} L_A \bar{L}_{\lambda^*} + \frac{1}{\mu} L_A \bar{L}_{\lambda^*}.$$

Proof. 1. From Lemma 2 we know that

$$\nabla \lambda_t^*(\mathbf{x}) = - \left[A [\nabla_{yy}^2 \phi_t^*]^{-1} A^T \right]^{-1} \left[A [\nabla_{yy}^2 \phi_t^*]^{-1} \nabla_{xy}^2 \phi_t^* \right]$$

To simplify the derivations we use the following notations for the components of $\nabla \lambda_t^*(\mathbf{x})$: $H(\mathbf{x}) := A [\nabla_{yy}^2 \phi_t^*]^{-1} A^T$, $G(\mathbf{x}) := [\nabla_{yy}^2 \phi_t^*]^{-1}$, $M(\mathbf{x}) = \nabla_{xy}^2 \phi_t^*$. Then we get

$$\begin{aligned}
& \|\nabla \lambda_t^*(\mathbf{x}) - \nabla \lambda_t^*(\bar{\mathbf{x}})\| \tag{15} \\
&= \|H^{-1}(\mathbf{x}) A G(\mathbf{x}) M(\mathbf{x}) - H^{-1}(\bar{\mathbf{x}}) A G(\bar{\mathbf{x}}) M(\bar{\mathbf{x}})\| \\
&\leq \|H^{-1}(\mathbf{x}) A G(\mathbf{x}) M(\mathbf{x}) - H^{-1}(\bar{\mathbf{x}}) A G(\mathbf{x}) M(\mathbf{x})\| \\
&+ \|H^{-1}(\bar{\mathbf{x}}) A G(\mathbf{x}) M(\mathbf{x}) - H^{-1}(\bar{\mathbf{x}}) A G(\bar{\mathbf{x}}) M(\bar{\mathbf{x}})\| \\
&\leq \|H^{-1}(\mathbf{x}) - H^{-1}(\bar{\mathbf{x}})\| \|A G(\mathbf{x}) M(\mathbf{x})\| \\
&+ \|H^{-1}(\bar{\mathbf{x}})\| \|A\| \|G(\mathbf{x}) M(\mathbf{x}) - G(\bar{\mathbf{x}}) M(\bar{\mathbf{x}})\| \\
&\leq \|H^{-1}(\mathbf{x})\| \|H(\mathbf{x}) - H(\bar{\mathbf{x}})\| \|H^{-1}(\bar{\mathbf{x}})\| \|A\| \|G(\mathbf{x})\| \|M(\mathbf{x})\| \\
&+ \|H^{-1}(\bar{\mathbf{x}})\| \|A\| \|G(\mathbf{x}) M(\mathbf{x}) - G(\bar{\mathbf{x}}) M(\bar{\mathbf{x}})\| \\
&+ \|G(\bar{\mathbf{x}}) M(\mathbf{x}) - G(\bar{\mathbf{x}}) M(\bar{\mathbf{x}})\| \\
&\leq \|H^{-1}(\mathbf{x})\| \|H(\mathbf{x}) - H(\bar{\mathbf{x}})\| \|H^{-1}(\bar{\mathbf{x}})\| \|A\| \|G(\mathbf{x})\| \|M(\mathbf{x})\| \\
&+ \|H^{-1}(\bar{\mathbf{x}})\| \|A\| \|G(\mathbf{x}) - G(\bar{\mathbf{x}})\| \|M(\mathbf{x})\| \\
&+ \|G(\bar{\mathbf{x}})\| \|M(\mathbf{x}) - M(\bar{\mathbf{x}})\|.
\end{aligned}$$

From Lemma 5 we see that $\|G(\mathbf{x})\| \leq 1/\mu$, $\|G(\bar{\mathbf{x}})\| \leq 1/\mu$, $\|M(\mathbf{x})\| \leq \bar{L}_{\phi xy}$, $\|H^{-1}(\mathbf{x})\| \leq \bar{L}_{A, \phi_t^{-1}}$, $\|H^{-1}(\bar{\mathbf{x}})\| \leq \bar{L}_{A, \phi_t^{-1}}$.

Also, it holds that

$$\begin{aligned}
\|M(\mathbf{x}) - M(\bar{\mathbf{x}})\| &= \|\nabla_{xy} \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_{xy} \phi_t(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))\| \\
&\leq L_{\phi xy} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|]
\end{aligned}$$

$$\begin{aligned}
\|G(\mathbf{x}) - G(\bar{\mathbf{x}})\| &= \left\| [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*)]^{-1} - [\nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)]^{-1} \right\| \\
&\leq \left\| [\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*)]^{-1} \right\| \left\| \nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*) \right\| \left\| [\nabla_{yy}^2 \phi_t(\bar{\mathbf{x}}, \bar{\mathbf{y}}_t^*)]^{-1} \right\| \\
&\leq \frac{L_{\phi yy}}{\mu^2} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|] \tag{16}
\end{aligned}$$

$$\begin{aligned}
\|H(\mathbf{x}) - H(\bar{\mathbf{x}})\| &= \|A \nabla_{yy}^{-1} \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) A^T - A \nabla_{yy}^{-1} \phi_t(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}})) A^T\| \\
&\leq \frac{L_A^2 L_{\phi yy}}{\mu^2} [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|]
\end{aligned}$$

Plugging the above into (15) results to

$$\begin{aligned}
& \|\nabla \lambda_t^*(\mathbf{x}) - \nabla \lambda_t^*(\bar{\mathbf{x}})\| \\
&\leq \left[\bar{L}_{A, \phi_t^{-1}} \frac{L_A^2 L_{\phi yy}}{\mu^2} \bar{L}_{A, \phi_t^{-1}} L_A \frac{1}{\mu} \bar{L}_{\phi xy} \right. \\
&+ \left. \bar{L}_{A, \phi_t^{-1}} L_A \left(\frac{L_{\phi yy}}{\mu^2} \bar{L}_{\phi xy} + \frac{1}{\mu} L_{\phi xy} \right) \right] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|].
\end{aligned}$$

We set

$$\begin{aligned}
L_{\lambda^*} &= \bar{L}_{A, \phi_t^{-1}} \frac{L_A^2 L_{\phi yy}}{\mu^2} \bar{L}_{A, \phi_t^{-1}} L_A \frac{1}{\mu} \bar{L}_{\phi xy} \\
&+ \bar{L}_{A, \phi_t^{-1}} L_A \left(\frac{L_{\phi yy}}{\mu^2} \bar{L}_{\phi xy} + \frac{1}{\mu} L_{\phi xy} \right).
\end{aligned}$$

The proof is complete.

2. From Lemma 2 we know that

$$\nabla \mathbf{y}_t^*(\mathbf{x}) = [\nabla_{yy}^2 \phi_t^*]^{-1} \left[-\nabla_{xy}^2 \phi_t^* - A^T \nabla \lambda_t^*(\mathbf{x}) \right].$$

We use the notation $\phi_t^* := \phi_t(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))$ and $\bar{\phi}_t^* := \phi_t(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))$ and obtain

$$\begin{aligned}
& \|\nabla \mathbf{y}_t^*(\mathbf{x}) - \nabla \mathbf{y}_t^*(\bar{\mathbf{x}})\| \\
&= \|[\nabla_{yy}^2 \phi_t^*]^{-1} \nabla_{xy}^2 \phi_t^* + [\nabla_{yy}^2 \phi_t^*]^{-1} A^T \nabla \lambda_t^*(\mathbf{x}) \\
&- [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} \nabla_{xy}^2 \bar{\phi}_t^* - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} A^T \nabla \lambda_t^*(\bar{\mathbf{x}})\| \\
&\stackrel{(a)}{\leq} \|[\nabla_{yy}^2 \phi_t^*]^{-1} \nabla_{xy}^2 \phi_t^* - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} \nabla_{xy}^2 \bar{\phi}_t^*\| \\
&+ \|[\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} \nabla_{xy}^2 \phi_t^* - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} \nabla_{xy}^2 \bar{\phi}_t^*\| \\
&+ \|[\nabla_{yy}^2 \phi_t^*]^{-1} A^T \nabla \lambda_t^*(\mathbf{x}) - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} A^T \nabla \lambda_t^*(\mathbf{x})\| \\
&+ \|[\nabla_{yy}^2 \phi_t^*]^{-1} A^T \nabla \lambda_t^*(\mathbf{x}) - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1} A^T \nabla \lambda_t^*(\bar{\mathbf{x}})\| \\
&\stackrel{(b)}{\leq} \|[\nabla_{yy}^2 \phi_t^*]^{-1} - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1}\| \|\nabla_{xy}^2 \phi_t^*\| \\
&+ \|[\nabla_{yy}^2 \bar{\phi}_t^*]^{-1}\| \|\nabla_{xy}^2 \phi_t^* - \nabla_{xy}^2 \bar{\phi}_t^*\| \\
&+ \|[\nabla_{yy}^2 \phi_t^*]^{-1} - [\nabla_{yy}^2 \bar{\phi}_t^*]^{-1}\| \|A\| \|\nabla \lambda_t^*(\mathbf{x})\| \\
&+ \|[\nabla_{yy}^2 \bar{\phi}_t^*]^{-1}\| \|A\| \|\nabla \lambda_t^*(\mathbf{x}) - \nabla \lambda_t^*(\bar{\mathbf{x}})\| \\
&\stackrel{(c)}{\leq} \left[\frac{\bar{L}_{\phi xy} L_{\phi yy}}{\mu^2} + \frac{L_{\phi xy}}{\mu} + \frac{L_{\phi yy}}{\mu^2} L_A \bar{L}_{\lambda^*} + \frac{1}{\mu} L_A \bar{L}_{\lambda^*} \right] \\
&\cdot [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^* - \bar{\mathbf{y}}_t^*\|],
\end{aligned}$$

where in (a) we add and subtract some terms and apply the triangle inequality; (c) results from applying eq. (16) and Lemma 5 in the first line of (b), eq. (16) and Lemma 7 in the second line of (b), eq. (16) and Lemma 6 in the third line of (b), eq. (16) and Lemma 8 in the fourth line of (b). We set $L_{y^*} = \frac{\bar{L}_{\phi_{xy}} \bar{L}_{\phi_{yy}}}{\mu^2} + \frac{\bar{L}_{\phi_{xy}}}{\mu} + \frac{\bar{L}_{\phi_{yy}}}{\mu^2} L_A \bar{L}_{\lambda^*} + \frac{1}{\mu} L_A \bar{L}_{\lambda^*}$. The proof is complete. $\square$

Proof of Lemma 3. In Lemma 2 we established that

$$\nabla F(\mathbf{x}) = \nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})).$$

Then,

$$\begin{aligned} & \|\nabla F(\mathbf{x}) - \nabla F(\bar{\mathbf{x}})\| \\ &= \|\nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \\ & \quad - \nabla_x f(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}})) - \nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))\| \\ &\leq \|\nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_x f(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))\| \\ & \quad + \|\nabla_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \\ & \quad + \|\nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))\| \\ &\stackrel{(a)}{\leq} L_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\|] \\ & \quad + \|\nabla_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \\ & \quad + \|\nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_{\mathbf{y}_t^*(\bar{\mathbf{x}})} \nabla_y f(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))\| \\ &\stackrel{(b)}{\leq} L_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\|] \\ & \quad + L_{y^*} \bar{L}_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\|] \\ & \quad + \bar{L}_{y^*} L_f [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\|] \\ &= [L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\|] \end{aligned} \quad (17)$$

where in (a) we used Ass. 2.1; in (b) we used Lemmas 6, 8 and Ass. 2.1, 2.5.

Next, notice that the bounds on Lemma 6 also hold for the individual vector components, i.e., $\|\nabla_{\mathbf{y}_{t,i}^*(\mathbf{x})}\| \leq \bar{L}_{y^*}, \forall \mathbf{x} \in \mathbb{R}^n, \forall i \in [K]$, where $\mathbf{y}_{t,i}^*(\mathbf{x})$ is the i th component of the vector $\mathbf{y}_t^*(\mathbf{x})$. This bounded gradient property implies the Lipschitz continuity of the function $\mathbf{y}_{t,i}^*(\mathbf{x})$, i.e., $\|\mathbf{y}_{t,i}^*(\mathbf{x}) - \mathbf{y}_{t,i}^*(\bar{\mathbf{x}})\| \leq \bar{L}_{y^*} \|\mathbf{x} - \bar{\mathbf{x}}\|, \forall \mathbf{x}, \bar{\mathbf{x}} \in \mathbb{R}^n, \forall i \in [K]$. Then, it follows that

$$\begin{aligned} \|\mathbf{y}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\bar{\mathbf{x}})\| &= \left[\sum_{i=1}^K (\mathbf{y}_{t,i}^*(\mathbf{x}) - \mathbf{y}_{t,i}^*(\bar{\mathbf{x}}))^2 \right]^{1/2} \\ &\leq \left[\sum_{i=1}^K \bar{L}_{y^*}^2 \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \right]^{1/2} \\ &= \left[K \bar{L}_{y^*}^2 \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \right]^{1/2} \\ &= \sqrt{K} \bar{L}_{y^*} \|\mathbf{x} - \bar{\mathbf{x}}\|. \end{aligned}$$

Combining the above expression with (17) results to

$$\begin{aligned} & \|\nabla F(\mathbf{x}) - \nabla F(\bar{\mathbf{x}})\| \\ &\leq [L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f] [\|\mathbf{x} - \bar{\mathbf{x}}\| + \sqrt{K} \bar{L}_{y^*} \|\mathbf{x} - \bar{\mathbf{x}}\|] \\ &\leq [L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f] \left(1 + \sqrt{K} \bar{L}_{y^*}\right) \|\mathbf{x} - \bar{\mathbf{x}}\|. \end{aligned}$$

We set $L_F := [L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f] \left(1 + \sqrt{K} \bar{L}_{y^*}\right)$. The proof is complete. $\square$

A.4. Proof of Lemma 4

We will prove that

$$\|\nabla F(\mathbf{x}) - \widehat{\nabla} F(\mathbf{x})\| \leq \widehat{L}_{F,t} \cdot \delta,$$

for some constant $\widehat{L}_{F,t}$ where

$$\begin{aligned} \nabla F(\mathbf{x}) &= \nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \\ \widehat{\nabla} F(\mathbf{x}) &= \nabla_x f(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x})) + \widehat{\nabla}_{\mathbf{y}_t^*(\mathbf{x})} \nabla_y f(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x})), \end{aligned}$$

and $\|\widehat{\mathbf{y}}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\mathbf{x})\| \leq \delta, \forall \mathbf{x} \in \mathbb{R}^n$. The expression $\widehat{\nabla}_{\mathbf{y}_t^*(\mathbf{x})}$ denotes the approximate implicit gradient, which we can obtain by substituting the exact solution $\mathbf{y}_t^*(\mathbf{x})$ with the approximate one $\widehat{\mathbf{y}}_t^*(\mathbf{x})$ in eq. (6), (7), (8); the approximate expression $\widehat{\nabla} \lambda_t^*(\mathbf{x})$ is defined similarly. Moreover, the proof of this Lemma is very similar to the proof of Lemma 3 and its prerequisites (Lemmas 5, 6, 7, 8). We explain in more details below.

First, in the context of Lemma 4 we assume that Ass. 3 also holds for the approximate solution $\widehat{\mathbf{y}}_t^*(\mathbf{x})$, i.e.,

$$-d_t \leq h_i(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x})) \leq -c_t < 0, \quad \forall \mathbf{x} \in \mathbb{R}^n, \forall i \in [K], \quad (18)$$

where c_t, d_t are defined in Ass. 3. Next, it can be shown that the Lemmas 5, 6, 7, 8 also hold when in the place of the exact solution $\mathbf{y}_t^*(\mathbf{x})$ we have the approximate one $\widehat{\mathbf{y}}_t^*(\mathbf{x})$. We rewrite the statements below for clarity.

Lemma 9. *The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$.*

1. $\nabla_{yy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x})) \succeq \mu I$
2. $\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x}))\|^{-1} \leq \frac{1}{\mu}$
3. $\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x}))\| \leq \bar{L}_{\phi_{yy}}$
4. $\|\nabla_{xy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x}))\| \leq \bar{L}_{\phi_{xy}}$
5. $\left\| \left[A \nabla_{yy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*(\mathbf{x})) \right]^{-1} A^T \right\|^{-1} \leq \bar{L}_{A, \phi_t^{-1}}$

where the constants $\bar{L}_{\phi_{yy}}, \bar{L}_{\phi_{xy}}, \bar{L}_{A, \phi_t^{-1}}$ are defined in Lemma 5.

Lemma 10. *The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$*

1. $\|\widehat{\nabla} \lambda_t^*(\mathbf{x})\| \leq \bar{L}_{\lambda^*}$
2. $\|\widehat{\nabla} \mathbf{y}_t^*(\mathbf{x})\| \leq \bar{L}_{y^*}$

where the constants $\bar{L}_{\lambda^*}, \bar{L}_{y^*}$ are defined in Lemma 6.

Lemma 11. *The following expressions hold for every $\mathbf{x} \in \mathbb{R}^n$*

1. $\|\nabla_{yy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{yy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*)\| \leq L_{\phi_{yy}} \|\mathbf{y}_t^*(\mathbf{x}) - \widehat{\mathbf{y}}_t^*(\mathbf{x})\|$
2. $\|\nabla_{xy}^2 \phi_t(\mathbf{x}, \mathbf{y}_t^*) - \nabla_{xy}^2 \phi_t(\mathbf{x}, \widehat{\mathbf{y}}_t^*)\| \leq L_{\phi_{xy}} \|\mathbf{y}_t^*(\mathbf{x}) - \widehat{\mathbf{y}}_t^*(\mathbf{x})\|$

where $\mathbf{y}_t^* = \mathbf{y}_t^*(\mathbf{x}), \widehat{\mathbf{y}}_t^* = \widehat{\mathbf{y}}_t^*(\mathbf{x})$ and the constants $L_{\phi_{yy}}, L_{\phi_{xy}}$ are defined in Lemma 7.

Lemma 12. *The following hold for every $\mathbf{x} \in \mathbb{R}^n$*

1. $\|\nabla \lambda_t^*(\mathbf{x}) - \widehat{\nabla} \lambda_t^*(\mathbf{x})\| \leq L_{\lambda^*} \|\mathbf{y}_t^*(\mathbf{x}) - \widehat{\mathbf{y}}_t^*(\mathbf{x})\|$
2. $\|\nabla \mathbf{y}_t^*(\mathbf{x}) - \widehat{\nabla} \mathbf{y}_t^*(\mathbf{x})\| \leq L_{y^*} \|\mathbf{y}_t^*(\mathbf{x}) - \widehat{\mathbf{y}}_t^*(\mathbf{x})\|$

where L_{λ^*}, L_{y^*} are defined in Lemma 8.

Proof of Lemmas 9, 10, 11, 12. The proofs of Lemmas 9, 10, 11, 12 follow directly from the proofs of Lemmas 5, 6, 7, 8, respectively. The only differences are on the points on which the functions/gradients/Hessians involved are evaluated. Specifically, the exact LL solution of problem (5) $\mathbf{y}_t^* = \mathbf{y}_t^*(\mathbf{x})$ (in Lemmas 5, 6) is substituted by the approximate one $\hat{\mathbf{y}}_t^*(\mathbf{x})$. Also, in the place of point $(\bar{\mathbf{x}}, \mathbf{y}_t^*(\bar{\mathbf{x}}))$ (in Lemmas 7, 8) we use the point $(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x}))$. Then, the proof steps are exactly the same as in Lemmas 5, 6, 7, 8. The proof is complete. $\square$

Proof of Lemma 4. Under Ass. 1-3, and the assumption in (18), we have

$$\begin{aligned}
& \|\nabla F(\mathbf{x}) - \hat{\nabla} F(\mathbf{x})\| \\
&= \|\nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) + \nabla \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) \\
&\quad - \nabla_x f(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x})) - \hat{\nabla} \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x}))\| \\
&\leq \|\nabla_x f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_x f(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x}))\| \\
&\quad + \|\nabla \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \hat{\nabla} \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \\
&\quad + \|\hat{\nabla} \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \hat{\nabla} \mathbf{y}_t^*(\mathbf{x}) \nabla_y f(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x}))\| \\
&\stackrel{(a)}{\leq} L_f \|\mathbf{y}_t^*(\mathbf{x}) - \hat{\mathbf{y}}_t^*(\mathbf{x})\| \\
&\quad + \|\nabla \mathbf{y}_t^*(\mathbf{x}) - \hat{\nabla} \mathbf{y}_t^*(\mathbf{x})\| \|\nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x}))\| \\
&\quad + \|\hat{\nabla} \mathbf{y}_t^*(\mathbf{x})\| \|\nabla_y f(\mathbf{x}, \mathbf{y}_t^*(\mathbf{x})) - \nabla_y f(\mathbf{x}, \hat{\mathbf{y}}_t^*(\mathbf{x}))\| \\
&\stackrel{(b)}{\leq} L_f \|\mathbf{y}_t^*(\mathbf{x}) - \hat{\mathbf{y}}_t^*(\mathbf{x})\| + L_{y^*} \bar{L}_f \|\mathbf{y}_t^*(\mathbf{x}) - \hat{\mathbf{y}}_t^*(\mathbf{x})\| \\
&\quad + \bar{L}_{y^*} L_f \|\mathbf{y}_t^*(\mathbf{x}) - \hat{\mathbf{y}}_t^*(\mathbf{x})\| \\
&\stackrel{(c)}{\leq} [L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f] \delta,
\end{aligned}$$

where in (a) we used Ass. 2.1 ;in (b) we used Lemmas 10, 12 and Ass. 2.1, 2.5; (c) follows from the bound on the solution accuracy of the LL problem $\|\hat{\mathbf{y}}_t^*(\mathbf{x}) - \mathbf{y}_t^*(\mathbf{x})\| \leq \delta$. We set $\hat{L}_{F,t} = L_f + L_{y^*} \bar{L}_f + \bar{L}_{y^*} L_f$. The proof is complete. $\square$

A.5. Proof of Theorem 1

Proof of Theorem 1. We use the following notation $F^r := F(\mathbf{x}^r)$ and $\hat{\nabla} F^r := \hat{\nabla} F(\mathbf{x}^r)$. From the Lipschitz gradient property of F it follows that

$$\begin{aligned}
F^{r+1} &\leq F^r - \alpha \langle \nabla F^r, \mathbf{x}^{r+1} - \mathbf{x}^r \rangle + \frac{L_{F,t}}{2} \|\mathbf{x}^{r+1} - \mathbf{x}^r\|^2 \\
F^{r+1} &\leq F^r - \alpha \langle \nabla F^r, \hat{\nabla} F^r \rangle + \frac{L_{F,t} \alpha^2}{2} \|\hat{\nabla} F^r\|^2 \quad (19)
\end{aligned}$$

The identity $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\mathbf{a}^T \mathbf{b}$ implies that

$$-\alpha \langle \nabla F^r, \hat{\nabla} F^r \rangle = \frac{\alpha}{2} \|\nabla F^r - \hat{\nabla} F^r\|^2 - \frac{\alpha}{2} \|\nabla F^r\|^2 - \frac{\alpha}{2} \|\hat{\nabla} F^r\|^2.$$

Plugging the above expression into (19) results to

$$\begin{aligned}
F^{r+1} &\leq F^r + \frac{\alpha}{2} \|\nabla F^r - \hat{\nabla} F^r\|^2 - \frac{\alpha}{2} \|\nabla F^r\|^2 - \frac{\alpha}{2} \|\hat{\nabla} F^r\|^2 \\
&\quad + \frac{L_{F,t} \alpha^2}{2} \|\hat{\nabla} F^r\|^2 \\
F^{r+1} &\leq F^r + \frac{\alpha}{2} \|\nabla F^r - \hat{\nabla} F^r\|^2 - \frac{\alpha}{2} \|\nabla F^r\|^2 \\
&\quad + \left(\frac{L_{F,t} \alpha^2}{2} - \frac{\alpha}{2} \right) \|\hat{\nabla} F^r - \nabla F^r + \nabla F^r\|^2
\end{aligned}$$

Further, assuming that $\frac{L_{F,t} \alpha^2}{2} - \frac{\alpha}{2} > 0$ we have

$$\begin{aligned}
F^{r+1} &\leq F^r + \frac{\alpha}{2} \|\nabla F^r - \hat{\nabla} F^r\|^2 - \frac{\alpha}{2} \|\nabla F^r\|^2 \\
&\quad + 2 \left(\frac{L_{F,t} \alpha^2}{2} - \frac{\alpha}{2} \right) \|\hat{\nabla} F^r - \nabla F^r\| + 2 \left(\frac{L_{F,t} \alpha^2}{2} - \frac{\alpha}{2} \right) \|\nabla F^r\|^2 \\
F^{r+1} &\leq F^r + \left(L_{F,t} \alpha^2 - \frac{\alpha}{2} \right) \|\nabla F^r - \hat{\nabla} F^r\|^2 + (L_{F,t} \alpha^2 - \frac{3\alpha}{2}) \|\nabla F^r\|^2 \\
&\quad + \left(\frac{3\alpha}{2} - L_{F,t} \alpha^2 \right) \|\nabla F^r\|^2 \leq F^r - F^{r+1} + \left(L_{F,t} \alpha^2 - \frac{\alpha}{2} \right) \|\nabla F^r - \hat{\nabla} F^r\|^2 \\
&\quad + \sum_{r=0}^{T-1} \left(\frac{3\alpha}{2} - L_{F,t} \alpha^2 \right) \|\nabla F^r\|^2 \leq F^0 - \underline{F} \\
&\quad + \sum_{r=0}^{T-1} \left(L_{F,t} \alpha^2 - \frac{\alpha}{2} \right) \|\nabla F^r - \hat{\nabla} F^r\|^2,
\end{aligned}$$

where $\underline{F}$ is a lower bound on F . Next, under the assumption that $\frac{3\alpha}{2} - L_{F,t} \alpha^2 > 0$ we get

$$\frac{1}{T} \sum_{r=0}^{T-1} \|\nabla F^r\|^2 \leq \frac{1}{T} \frac{F^0 - F^T}{\frac{3\alpha}{2} - L_{F,t} \alpha^2} + \frac{L_{F,t} \alpha^2 - \frac{\alpha}{2}}{\frac{3\alpha}{2} - L_{F,t} \alpha^2} \hat{L}_F^2$$

We set $\alpha = \frac{5}{4L_{F,t}}$ in the above expression,

$$\frac{1}{T} \sum_{r=0}^{T-1} \|\nabla F(\mathbf{x}^r)\|^2 \leq \frac{16}{5T} (F(\mathbf{x}^0) - \underline{F}) L_{F,t} + 3\hat{L}_{F,t} \cdot \delta^2,$$

where the constants $L_{F,t}, \hat{L}_{F,t}$ are defined in Lemmas 3 and 4, respectively. The proof is complete. $\square$